HAUSDORFF DIMENSION N-2 OF THE SYMMETRIC BOUNDARY OF A STRICTLY CONVEX PROJECTIVE DOMAIN WITH C1 BOUNDARY

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Editorial note. The notation x* for the tangent hyperplane at a point of the boundary, used by the statement, is introduced in the context paragraph at source lines 1003-1005 and again inside the statement's own display; both are authors' text.

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Allowed standard prerequisites (editorial).

- Hausdorff dimension and Hausdorff measure of a subset of a Euclidean space. DimH(A) is the infimum of the s for which the s-dimensional Hausdorff measure of A vanishes; a bi-Lipschitz image of A has the same dimension, and a graph of a Lipschitz function on an open subset of R^m has dimension m.. the authors' own definition and conventions at source lines 309-331. It is the invariant being computed and the tool used at the end of the proof, where the set is shown to be a 1-Lipschitz graph over a spacelike subspace of dimension n-2.

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- Duality and affine charts in real projective space. Every point x of a convex domain boundary has a tangent hyperplane x*, the dual projective space parametrizes hyperplanes, and after choosing two boundary points x_0, x_infinity with covectors alpha_0, alpha_infinity the affine patch P(R^n) minus x_infinity^* is identified with V x R, V = x_0^* cap x_infinity^*, by (v,t) -> [v + t v_infinity + v_0].. the authors' own setup at source lines 1017-1023. Only this elementary projective duality and the two explicit affine charts are assumed; the formula for the dual point and the dimension bound are retained obligations.
- Strict convexity of a C^1 function and Brouwer's invariance of domain. For a strictly convex C^1 function f one has d_u f(v-u) < f(v) - f(u) for u different from v; an injective continuous map from an open subset of R^m into R^m is open.. used at source lines 1055-1060. Strict convexity supplies the strict positivity of the quadratic form on difference vectors, and invariance of domain converts injectivity of the orthogonal projection onto a spacelike subspace into the graph representation that yields the 1-Lipschitz bound.

A particular case of strongly projectively convex-cocompact group in $\mathrm{SL}(n,\mathbb{R})$ is a group acting properly discontinuously and cocompactly on a Gromov-hyperbolic convex set. Crampon proved that the Hilbert critical exponent of such a group is at most $n-2$. we prove here that the Hausdorff dimension of $\Lambda_\Gamma^{\mathrm{sym}}$ equals $n-2$, so that our main theorem does recover Crampon's result.

If Γ acts properly discontinuously and cocompactly on a Gromov hyperbolic convex domain Ω, then we have $\Lambda_\Gamma = \partial\Omega$, $\Lambda_\Gamma^* = \partial\Omega^*$ (the set of hyperplanes tangent to $\partial\Omega$) and $\Lambda_\Gamma^{\mathrm{sym}} \subset \Lambda_\Gamma \times \Lambda_\Gamma^*$ is the graph of the homeomorphism mapping $x \in \partial\Omega$ to $x^* := T_x\partial\Omega$.

Theorem 2.55. *Let Ω be a strictly convex domain of $\mathbf{P}(\mathbb{R}^n)$ with $\mathcal{C}^1$ boundary. Then*

$$\partial\Omega^{\mathrm{sym}} := \{(x, x^*) \mid x \in \partial\Omega\} \subset \partial\Omega \times \partial\Omega^*$$

has Hausdorff dimension $n-2$.

Note that, while $\partial\Omega$ and $\partial\Omega^*$ are $\mathcal{C}^1$ spheres of dimension $n-2$, the homeomorphism $x \mapsto x^*$ is not $\mathcal{C}^1$ unless $\partial\Omega$ is $\mathcal{C}^2$ (which is very constraining for divisible convex sets). It is thus not immediate that its graph has Hausdorff dimension $n-2$ in general.

Proof of Theorem 2.55. Fix two distinct points $x_0, x_\infty \in \partial\Omega$, and consider vectors $v_0 \in x_0$, $v_\infty \in x_\infty$ and linear forms $\alpha_0 \in x_0^*$, $\alpha_\infty \in x_\infty^*$ such that $\alpha_0(v_\infty) = \alpha_\infty(v_0) = 1$. Let $V = x_0^* \cap x_\infty^* \subset \mathbb{R}^n$, so that the affine patch $\mathbf{P}(\mathbb{R}^n) \setminus x_\infty^*$ can be identified with $V \times \mathbb{R}$ through the map $(v,t) \mapsto [v + tv_\infty + v_0]$.

Since Ω is strictly convex and has $\mathcal{C}^1$ boundary, under this identification $\partial\Omega \setminus \{x_\infty\}$ is the graph of a $\mathcal{C}^1$ strictly convex function $f : V \to \mathbb{R}$.

Let $p_{0,\infty} : \mathbb{R}^n \to V$ be the projection with kernel $x_0 \oplus x_\infty$, so that we also get an identification of the affine patch $\mathbf{P}\left((\mathbb{R}^n)^*\right) \setminus x_0^\perp$ with $V^* \times \mathbb{R}$ through the map $(\alpha, t) \mapsto [\alpha \circ p_{0,\infty} + t\alpha_\infty + \alpha_0]$.

Under this identification, the dual point of $(u, f(u)) \in \partial\Omega \setminus \{x_\infty\}$ is

$$(u, f(u))^* = (-d_u f, d_u f(u) - f(u)) \in \partial\Omega^* \setminus \{x_\infty^*\}.$$

It follows that the map

$$\begin{aligned} \partial\Omega^* \setminus \{x_\infty^*\} &\to V^* \\ (u, f(u))^* &\longmapsto -d_u f \end{aligned}$$

is just the projection on the first coordinate of a $\mathcal{C}^1$ submanifold of $V^* \times \mathbb{R}$, so it is $\mathcal{C}^1$ (even though it is the composition of the non $\mathcal{C}^1$ maps $x^* \mapsto x$ and $v \mapsto d_v f$).

The projection

$$\begin{aligned}\pi : (\partial\Omega \setminus \{x_\infty\}) \times (\partial\Omega^* \setminus \{x^*_\infty\}) &\to V \times V^* \\ ((u, f(u)), (v, f(v))^*) &\longmapsto (u, \mathrm{d}_v f)\end{aligned}$$

is a $\mathcal{C}^1$ isomorphism. It is thus enough to prove that

$$\pi(\partial\Omega^{\mathrm{sym}} \setminus \{(y, y^*)\}) = \{(u, \mathrm{d}_u f) \mid u \in V\}$$

has Hausdorff dimension $d - 2$.

Now let us equip $V \times V^*$ with the symmetric bilinear form

$$\langle (u, \alpha), (v, \beta) \rangle = \alpha(v) + \beta(u).$$

This form is non-degenerate of signature $(n-2, n-2)$, and we claim that

$$\pi(\partial\Omega^{\mathrm{sym}} \setminus \{(y, y^*)\})$$

is a spacelike topological submanifold of dimension $n-2$, hence a Lipschitz manifold.

Indeed, for every $u \neq v \in V$, we have

$$\big\langle (v-u, \mathrm{d}_v f - \mathrm{d}_u f), (v-u, \mathrm{d}_v f - \mathrm{d}_u f) \big\rangle = -2(\mathrm{d}_u f(v-u) + \mathrm{d}_v f(u-v)).$$

By strict convexity of f, we have

$$\mathrm{d}_u f(v-u) < f(v) - f(u) \quad \text{and} \quad \mathrm{d}_v f(u-v) < f(u) - f(v),$$

from which we get

$$\big\langle (v-u, \mathrm{d}_v f - \mathrm{d}_u f), (v-u, \mathrm{d}_v f - \mathrm{d}_u f) \big\rangle > 0.$$

Now, let W be a spacelike subspace of $V \times V^*$ of dimension $n-2$. The previous computation implies that the orthogonal projection of $\pi(\partial\Omega^{\mathrm{sym}} \setminus \{(x_\infty, x^*_\infty)\})$ to W is injective. Since $\pi(\partial\Omega^{\mathrm{sym}} \setminus \{(x_\infty, x^*_\infty)\})$ is a $\mathcal{C}^1$ manifold of dimension $n-2$, this projection is open (by Brouwer's invariance of domain), and $\pi(\partial\Omega^{\mathrm{sym}} \setminus \{(x_\infty, x^*_\infty)\})$ is the graph of a function from an open domain of W to $W^\perp$. Finally, since $\pi(\partial\Omega^{\mathrm{sym}} \setminus \{(x_\infty, x^*_\infty)\})$ is spacelike, this graph is 1-Lipschitz (when W is endowed with $\langle \cdot, \cdot \rangle$ and $W^\perp$ with $-\langle \cdot, \cdot \rangle$). We thus conclude that $\pi(\partial\Omega^{\mathrm{sym}} \setminus \{(x_\infty, x^*_\infty)\})$ is a Lipschitz manifold of dimension $n-2$, hence $\partial\Omega^{\mathrm{sym}}$ has Hausdorff dimension $n-2$. □